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Rational decisions in multi-step environments with few rollouts

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Abstract

People routinely make decisions by mentally simulating the potential outcomes of their actions. However, this process appears computationally intractable in real-world situations involving sequences of decisions with exponentially many possible futures. How the brain efficiently evaluates temporally extended decisions despite limited cognitive resources remains a fundamental puzzle. Here we present a mathematical theory showing that for decisions in multi-step environments, the rational strategy is to perform only a few mental simulations, formalized as rollouts. This is because the first rollouts provide substantially more information than later ones despite taking a similar amount of time, so the opportunity cost of additional simulations quickly outweighs their marginal benefit. Our framework demonstrates that this efficiency relies on the correlated reward structure of naturalistic environments, which allows information from one rollout to generalize to many related future paths. This theory also explains why, under resource constraints, many shallow rollouts are preferable to fewer deep ones; why apparently myopic decisions can arise without explicit temporal discounting; and how to relate the dynamics of planning to evidence accumulation models. We validate predictions of our theory in two behavioral experiments, which confirm that humans achieve higher reward rates with few rollouts and dynamically adjust their simulation depth based on available cognitive resources. These findings reveal how the brain balances the depth and breadth of mental simulation to make effective decisions under computational constraints, providing a unifying account of planning that bridges computationally intensive search algorithms in machine learning and the remarkable efficiency of human decision-making.

Keywords: decision-making; mental simulation; reinforcement learning; planning

1 Introduction

Mental simulation enables humans to evaluate potential action outcomes before committing to decisions. In both simple choices with immediate rewards and complex scenarios with long-term consequences, people mentally simulate future outcomes to guide their behavior (Battaglia et al., 2013; Mattar and Lengyel, 2022). This ability appears fundamental to intelligent decision-making, yet also intractable in real-world situations involving sequences of decisions with exponentially many possible futures. Thus, a central question in cognitive science is: how can mental simulation support effective and timely decision-making despite cognitive constraints?

In simple decision problems, this question has been addressed by conceptualizing decision-making as a process of sampling possible outcomes. Whether drawn from the external world or from memory, these samples allow an agent to evaluate options and make good choices when determining the optimal solution is intractable. Influential theories of decision-making, such as drift-diffusion models, propose that people sample evidence until a decision threshold is met (Ratcliff and Smith, 2004; Ratcliff and McKoon, 2008; Gold and Shadlen, 2001; Busemeyer and Townsend, 1993). While such stochastic sampling can potentially require many samples for accurate decisions, empirical studies consistently show that people use relative few samples to guide their choices (Erev and Roth, 2014; Hertwig et al., 2004; Hertwig and Erev, 2009). This surprising observation is explained by another influential theory demonstrating that, when samples carry opportunity costs, deciding based on just one or very few mental simulations is not only possible, but the *optimal* strategy (Vul et al., 2014). This insight provides a normative foundation for understanding how mental simulation can be both minimal and effective, at least in simple bandit settings where actions yield immediate, one-time outcomes.

Real-world decisions, however, rarely resemble simple bandit problems. Most consequential choices — accepting a job offer, choosing an investment, or selecting a medical treatment — result in a cascade of events that unfold stochastically over time. In such multi-step environments, mental simulation requires navigating a complex decision tree where the number of possible futures grows exponentially with the depth of the plan. Densely simulating such a tree is not a reasonable strategy given the brain's limited time and computational power. Although humans employ various heuristics to reduce this computational burden, such as pruning unpromising action action sequences (Huys et al., 2012) and caching the results of prior computations (Huys et al., 2015), whether and how mental simulation supports decision-making in such vast state spaces remains an open question.

In this study, we demonstrate that the optimal number of mental simulations in multi-step environments is, contrary to our intuition, still remarkably low. Through analytical derivations and simulations, we show that this efficiency has a different explanation than the one provided for simple choices. Here, it results from the correlated reward structure of the environment, which reduces the marginal benefit of additional sampling. Since mental simulations take time, the opportunity cost of additional simulations quickly surpass their marginal benefit. Thus, the optimal policy is to decide based on only a few simulations. Our theory further predicts that when computational resources are limited, shallower rollouts are optimal, whereas deeper rollouts become preferable only when resources are abundant.

We validate these theoretical predictions through behavioral experiments showing that humans achieve higher reward rates with few rollouts and adaptively adjust their simulation depth based on available resources. Our framework also explains several puzzling features of human planning: how myopic choices emerge from rational evaluation without explicit temporal discounting, and why evidence accumulation is slower in deeper decision trees. These findings reveal how correlated reward structures in naturalistic environments enable efficient generalization from minimal samples, bridging the gap between computationally intensive search algorithms in machine learning and the remarkable efficiency of human decision-making.

2 Results

2.1 Using rollouts to inform decisions in multi-step environments

Our study considers any task where an agent takes actions that yield rewards in a multi-step environment, applicable to a wide range of real-world scenarios. The agent's goal is to select an action that maximizes its expected cumulative future reward, where the action triggers a series of state transitions, with one reward in each state. We assume that only the first action is optimized, and the remaining transitions are uncontrollable or not optimized Zhou et al. (2024). This framework is relevant to many real-world situations where an initial decision sets off a chain of events beyond our control or where high uncertainty makes it impractical to optimize every future step. For instance, investing in the stock market involves choosing a portfolio today, with the final return being determined by subsequent market fluctuations outside the investor's control. Although these future events are uncontrollable, evaluating their potential risks and benefits is still essential.

The agent is equipped with a world model consisting of ground-truth reward and transition probabilities. To decide on an action, the agent iteratively considers possible future states and rewards that can be gained there, constructing a decision tree where nodes represent possible future states of the world and edges represent possible transitions (Figure 1A). We assume that the agent performs a finite number of forward mental simulations, formalized as "rollouts". Each rollout is a sampled trajectory from the root to a downstream node that provides a noisy estimate of the action's expected cumulative reward. We assume a simulation policy that samples all root-node actions evenly. After simulation, the agent selects the action with the highest average cumulative reward estimated from the rollouts. In general, performing more rollouts leads to more accurate value estimates and, consequently, better decision-making performance.

A key property of the decision tree is its hierarchical structure, where paths ending in nearby nodes tend to share more intermediate nodes (Figure 1A). We formalize this overlap using the most recent common ancestor (MRCA), s_{ij}^{MRCA}, which we define as the deepest node shared between the paths ending in nodes s_i and s_j. The depth of the MRCA d_{ij}^{MRCA} quantifies this path overlap; deeper MRCAs indicate more shared nodes, mirroring the tree's structure where closer leaves are more related (Figure 1B). This structural hierarchy leads to a corresponding hierarchy in the cumulative rewards, where paths with more shared nodes have more similar cumulative rewards. Consequently, the reward covariance matrix has the same structure as the MRCA depth matrix (Figure 1C). This correlation allows for powerful generalization, as a single rollout informs not only its own path but also other related paths that share common nodes.

To quantify this generalization, we consider a scenario in which a single rollout to a leaf node s_i yields a cumulative reward sample g_i. The agent can leverage this information to update its belief about expected cumulative reward m_j in any other path as follows:

$$\Delta\mathbb{E}[m_j] \propto d_{ij}^{\text{MRCA}} g_i, \tag{1}$$

$$\Delta\text{Var}[m_j] \propto -(d_{ij}^{\text{MRCA}})^2. \tag{2}$$

i.e., the change in the estimated mean cumulative reward is proportional to the path overlap, while the reduction in uncertainty (variance) depends quadratically on the overlap. Thus, a deeper MRCA results in greater information gain and stronger generalization. Although the exact updates become more complex after multiple rollouts, the fundamental principle holds: path overlap governs generalization (Figure 1DE).

The hierarchical structure of the decision tree effectively reduces the complexity of the decision problem. Although the number of paths grows exponentially with depth, the shared structure allows an agent to generalize across paths, reducing the number of rollouts needed for an accurate value estimate. Such generalization is not possible in bandit problems, where each option must be sampled independently. In multi-step tasks, it contributes to efficient decision-making even under limited time and cognitive resources.

2.2 Optimal decisions based on few rollouts

To investigate how people make decisions with long-term consequences, we designed a task requiring participants to interact with a visually displayed decision tree (Figure 2A). Nodes were arranged in a circular layout, each colored by its reward, with the root node colored red and arrows indicating transitions. The transition structure and reward at each node of the tree was randomized on each trial to ensure that participants had to rely on decision-time planning rather than memorization. In each trial, the participant's goal was to maximize their cumulative reward by selecting a depth-1 node. This choice triggered a sequence of random transitions to a leaf node, with rewards being collected along the way (Figure 2C). In the task, the rewards and transitions were not revealed upfront. Participants had to gather information by hovering their mouse over individual nodes (Figure 2B). Each hover revealed that node's reward and its outgoing transitions. Hovering was only allowed over the two children of the currently hovered node, or over the root node. This setup mimics rollout-based information sampling before decision-making.

Before analyzing human behavior, we examined the theoretically optimal number of rollouts in a task with a branching factor of $b = 2$ (that is, two children were available from each state). We operationalized performance as normalized accuracy — the probability of selecting the best root-node action, rescaled to the range between -1 and 1, and with chance accuracy being 0. In our simulations, we found that performance increases with the number of rollouts for trees of all depths (Figure 3A). This is because each rollout reduces uncertainty about the best choice.

However, rollouts take time in realistic scenarios, creating opportunity costs (Jensen et al., 2023; Agrawal et al., 2022): if simulating and acting must both happen in a limited amount of time, more time allocated to sampling implies less time acting. To formalize this trade-off, we defined performance per time as normalized accuracy divided by time needed for all rollouts plus one action. Our simulations show that, for a modest simulation cost of 0.1 (i.e., one action takes as much time

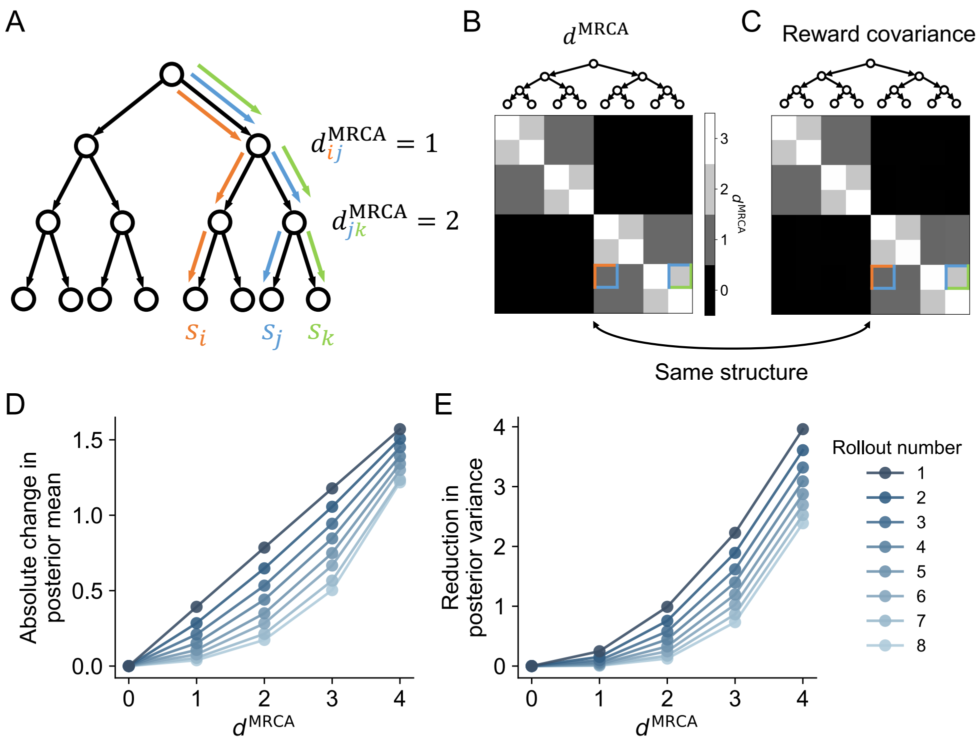

Figure 1: Temporally extended decision-making with rollouts. (A) A decision tree with branching factor $b = 2$. The highlighted trajectories show rollouts to three different leaf nodes, s_i, s_j, and s_k, respectively. The depth of the most recent common ancestor (MRCA) is $d_{ij}^{\mathrm{MRCA}} = 1$ between s_i and s_j, and $d_{jk}^{\mathrm{MRCA}} = 2$ between s_j and s_k. (B) The depth of the MRCA between two arbitrary paths. The corresponding tree structure is shown on top of the matrix. The colored squares indicate the depth of the MRCA between s_i and s_j, and between s_j and s_k shown in (A). (C) Covariance matrix of the cumulative reward between different paths in a decision tree. (D-E) The absolute change in posterior mean (D) and the reduction in posterior variance (E) resulting from a single rollout as a function of the depth of the MRCA. Colors indicate different rollout numbers.

as 10 simulations), performance per time peaks at a small number of rollouts (Figure 3B). Beyond this peak, the marginal benefit of additional sampling is outweighed by its cost. This finding holds across a range of tree depths and simulation costs; the optimal number of rollouts is generally small, only becoming large when simulations are very cheap and the decision tree is deep (Figure 3C). This pattern also persists for different branching factors (Figure S1). These results suggest that deciding after just a few rollouts is the optimal strategy.

To test these predictions experimentally, we conducted a behavioral experiment where participants could sample rollouts by hovering over tree nodes before selecting a depth-1 node with a click. The duration of the rollout phase varied across trials, and the trial-specific time limit was unknown to the participants. Consistent with model predictions, participants' performance improved with more rollouts (Figure 3D). Assuming equal time costs for rollouts and actions, participants' performance per time peaked at just two rollouts (Figure 3E). These results support our model predictions and the idea that a normative agent should make a decision based on very few samples.

While these findings appear similar to those from simpler bandit tasks, the underlying mechanism is different. In multi-step environments, efficiency stems from the reward correlation structure. This structure allows information from one rollout to generalize to many related future paths. To illustrate this, consider an environment with rewards only at terminal states, eliminating any path correlation. Here, a rollout along one path provides no information about other paths. To collect information about all paths, the agent must sample at least one rollout per path. In other words, a decision tree with branching factor b and depth d is equivalent to a bandit problem with b^d independent arms, whose resource requirements grow exponentially with tree depth.

We verified these predictions in correlated versus uncorrelated environments of varying configurations. These simulations confirm that, in correlated environments, the optimal number of rollouts grows only linearly with tree depth, and performance per time remains robust (Figure S2). In uncorrelated environments, however, the optimal number of rollouts increased

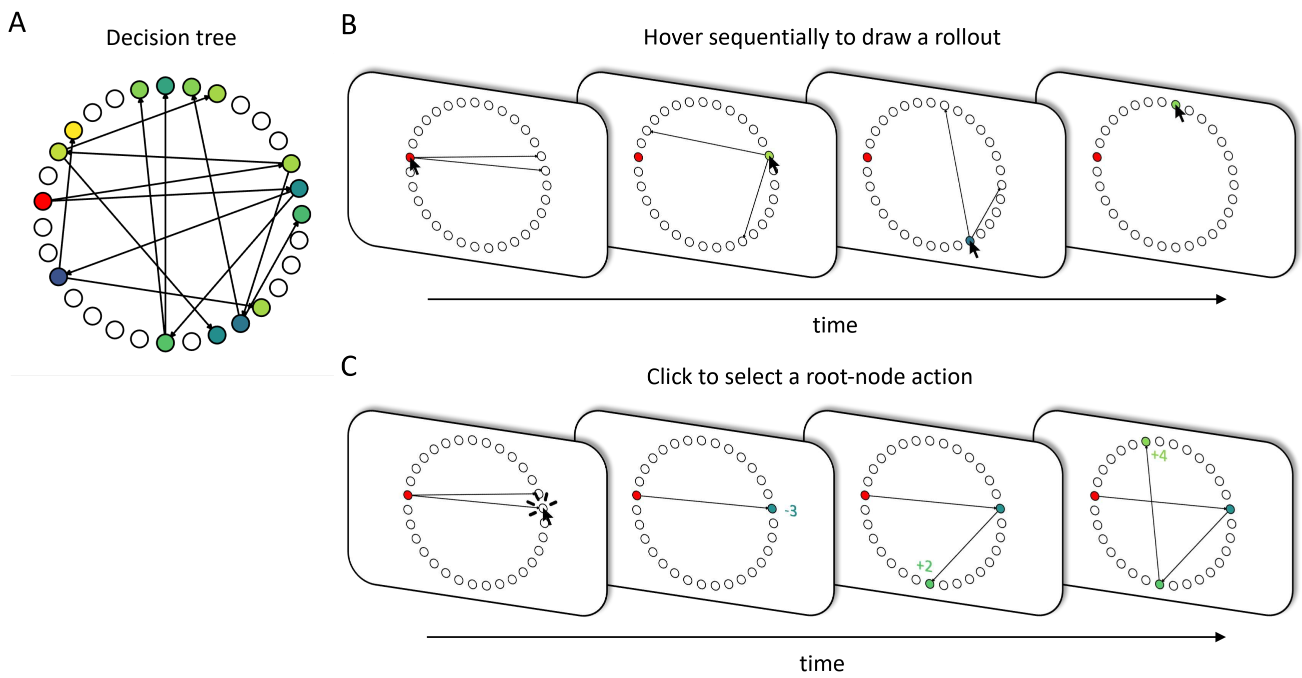

Figure 2: Task illustration. (A) The decision tree used in the experiments. Nodes were arranged on a circle. Each node was associated with a reward, indicated by its color, with yellow representing high rewards and blue representing low rewards. The root node was highlighted in red and transitions were indicated by arrows. (B) The layout shown to participants during a rollout. Participants moved their mouse to hover sequentially over nodes to sample a rollout. The reward associated with the hovered node and its outgoing transitions were revealed. (C) The time course of a decision. Participants clicked to choose a depth-1 node, and the rest of the trajectory was sampled randomly. Colored numbers were shown to indicate the empirical reward to the participant.

drastically, with the optimal performance per time dropping quickly with increasing tree depth or branching factor (Figure S2). These findings highlight the importance of structural correlations in supporting sample-efficient decisions. This principle is reflected in real-world choices where early decisions lead to similar futures; for instance, career paths following a medical degree are more related to each other than to those resulting from a law degree.

2.3 Shallow rollouts are optimal under resource constraints

So far, our analyses have assumed that rollouts traverse the full depth of the tree, but this strategy may not be optimal under resource constraints. Because the likelihood of visiting a specific node decreases with its depth, shallower nodes are more likely to be visited and are thus more informative for the agent's choice. To formalize the trade-off between exploring deep and shallow nodes, we endowed an agent with a finite "computational budget", defined as the total number of node expansions (i.e., simulations) it could perform before acting. This constraint creates a critical trade-off: with a fixed budget, the agent can perform either many shallow rollouts or fewer deep ones (Figure 4A-B).

Our model predicts that the optimal rollout depth is not fixed, but instead depends on the available computational budget. As expected, overall performance improved with a larger budget. This is because a higher budget allowed for more extensive sampling (Figure 4C). Critically, we found that shallow rollouts were superior under a small budget, as they efficiently reduce uncertainty about early nodes that have the strongest influence on the cumulative reward. As the budget increased, however, the marginal utility of further shallow rollouts diminished, and deeper rollouts became advantageous. This is because deeper simulations provide unique information about long-term outcomes that can differentiate between options with otherwise similar early paths. Furthermore, we found that the optimal rollout depth decreases as the tree's branching factor increases, since deeper nodes become quadratically harder to reach (Figure 4D). Taken together, our theory provides a rational explanation for why humans often constrain their simulation depth (Snider et al., 2015; Van Opheusden et al., 2023; Sezener et al., 2019): depth-limited planning strikes an optimal trade-off between simulation cost and informativeness.

To test whether humans adaptively adjust their planning depth in line with these normative predictions, we conducted a second experiment. In this experiment, we gave participants an "hover budget" on each trial. Participants were explicitly

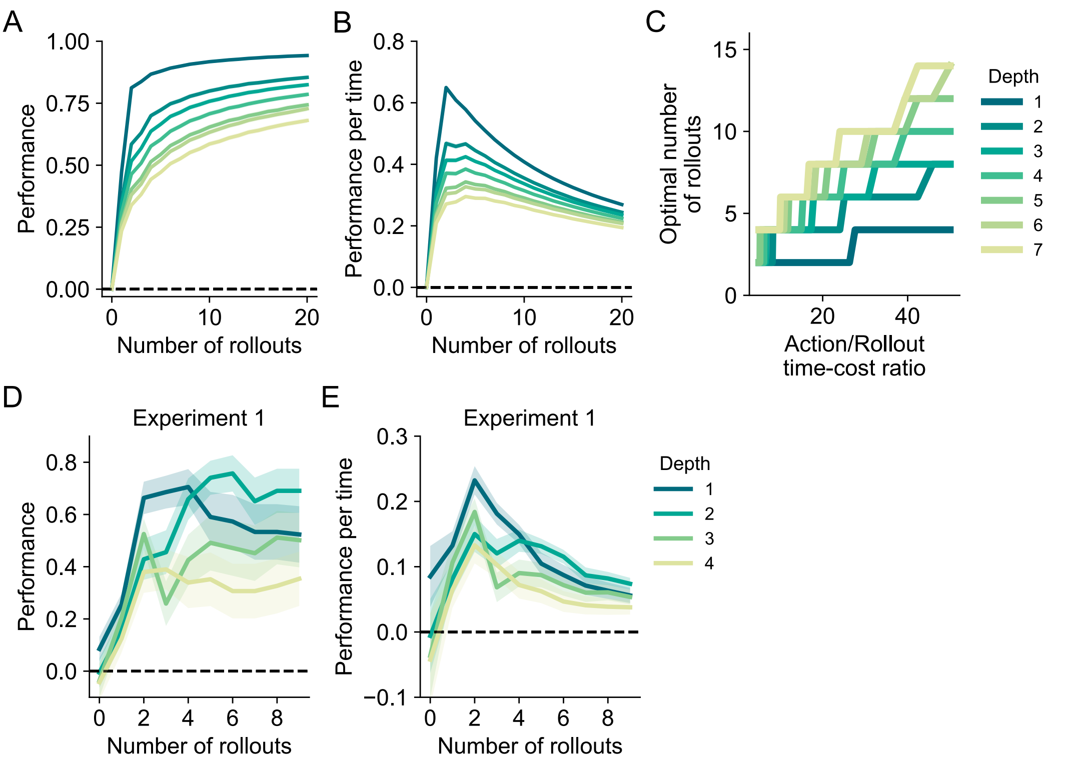

Figure 3: Optimal decisions require just a few rollouts. (A) Model performance as a function of the number of rollouts sample before each action. (B) Model performance per time, shown as a function of the number of rollouts performed before taking each action. (C) Optimal number of rollouts that yield the highest performance per time as a function of the action/rollout time-cost ratio. (D-E) Experimental data reproducing the simulation results from (A) and (B). Dashed lines indicate chance level performance. Shaded regions indicate standard errors across participants.

told what the budget was, which allowed them to adapt their rollout strategy according to the available resources. Consistent with our model, participants' average rollout depth systematically increased with larger budgets. Additionally, their planning was shallower in trees with higher branching factors, again matching our model's predictions (Figure 4E).

Our finding that resource limits favor shallow rollouts might seem to contradict prior work advocating that deep simulation is generally advantageous (Mastrogiuseppe and Moreno-Bote, 2022), but this apparent discrepancy is resolved by key differences in model assumptions. First, the model in Mastrogiuseppe and Moreno-Bote (2022) assumes deterministic rewards, where sampling a node once reveals its true value. This makes resampling a node useless and encourages exploration of new, deeper parts of the tree. In contrast, our environment has noisy rewards, making it advantageous to repeatedly sample shallower, high-probability nodes to reduce their uncertainty. Our setting is therefore more aligned with that of Moreno-Bote et al. (2020), who similarly found that resource limits favor breadth over depth. Second, the prior work assumes deterministic transitions, which allows an agent to reliably reach any promising node it discovers. This incentivizes deep simulations to find high-value distant rewards. Our model, however, uses stochastic transitions, which makes deeper nodes harder to reach and naturally discounts their value, prioritizing information about earlier rewards. Our framework is thus consistent with recent findings that planning depth decreases as environmental stochasticity increases (Lei et al., 2025).

2.4 Rollouts under uncertainty lead to temporal discounting

A hallmark of human decision-making is a preference for small, immediate rewards over larger, delayed ones, a phenomenon known as temporal discounting (Frederick et al., 2002; Mazur, 2013; Laibson, 1997; Green and Myerson, 2004). In sequential tasks, a related observation is that people are often averse to choices that involve early punishments, even if they lead to greater long-term rewards (Huys et al., 2012). Here, we show that our rollout-based model can naturally account for such myopic preferences without assuming an explicit devaluation of future outcomes, but rather as an emergent property of rational evaluation under uncertainty with shallow rollouts.

To test whether our rollout-based model shows myopic preferences, we modeled a scenario adapted from Huys et al. (2012) where an agent must choose between a safe, low-variance option and a risky, high-variance path involving an early

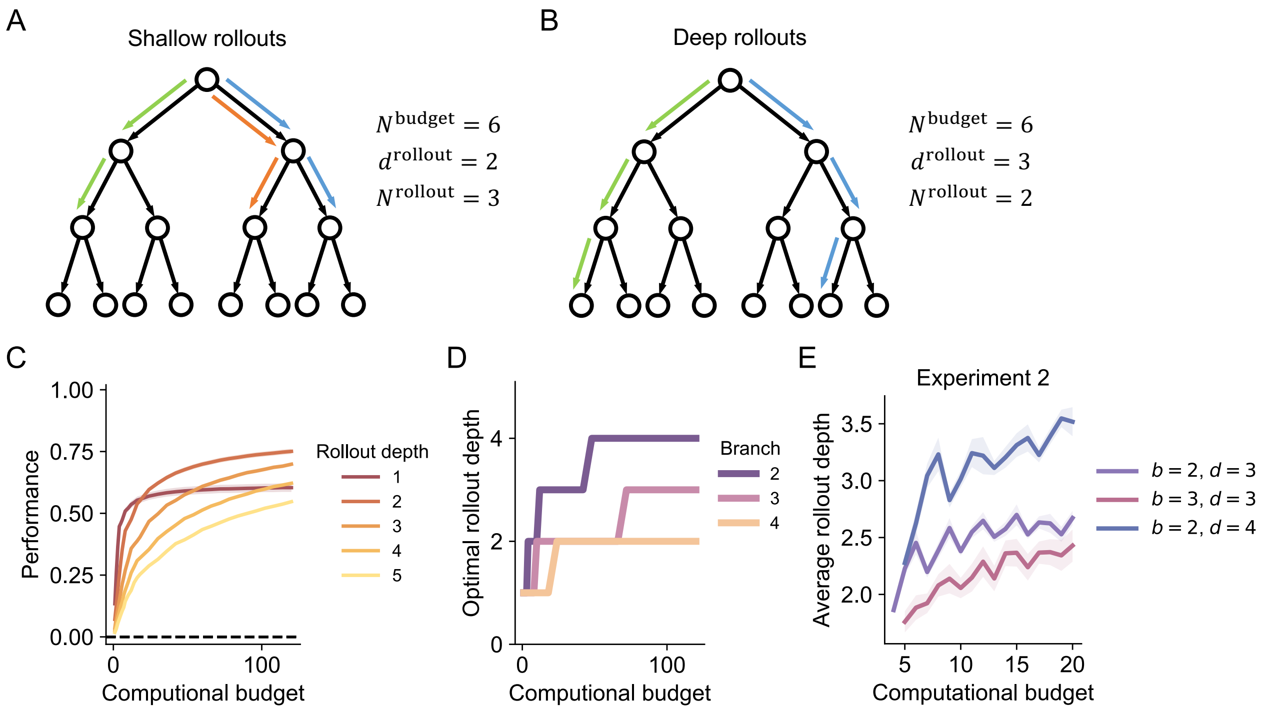

Figure 4: Optimal rollout depth varies with computational budget. (A-B) Illustration of shallow and deep rollouts. Given a fixed computational budget of 6 node expansions, the agent can choose to either (A) perform 3 rollouts, each having a depth of 2, or (B) 2 rollouts, each having a depth of 3. (C) Model performance as a function of computational budget (the number of node expansions before taking an action) under different rollout depths and with a branching factor of $b = 3$. Dashed lines indicate chance level performance. (D) Optimal rollout depth as a function of computational budget for different branching factors. Rollout depth is rounded to ensure the total number of node expansions remains within the budget limit. (E) Average rollout depth as a function of computational budget in human participants. b and d are the branching factor and tree depth used in the experiment. Shaded regions show standard errors across participants.

punishment followed by a larger, delayed reward. Specifically, we configured one action to lead to a punishment of -10 at depth 1, followed by a reward of $+15$ at a designated downstream leaf node, while all other nodes contained rewards sampled randomly from our standard distribution (Figure 5A). For this simulation, we assumed that environment transitions are deterministic and that the agent has full control over its policy at every node. Under these conditions, the high-variance action objectively maximizes expected cumulative reward. However, if there is uncertainty in the agent's internal world model, the high-variance action may be disfavored. This is because uncertainty can cause the agent to overestimate the risk of not getting the later pay-off after the initial large punishment. We modeled this uncertainty with a parameter ϵ representing the perceived probability that each intended transition toward the large reward might fail. This formulation captures a key aspect of real-world planning: even when agents intend to pursue a delayed reward and the environment itself is deterministic, internal uncertainty about successfully navigating to that reward can rationally discourage them from accepting early costs.

The agent's choices revealed that uncertainty drives temporal discounting in a depth-dependent manner. For small values of ϵ (low uncertainty), the agent correctly favored the high-variance action that maximizes expected reward. However, as ϵ increased, the agent increasingly favored the low-variance option, effectively avoiding the early punishment (Figure 5B). This discounting effect intensified with tree depth: deeper trees showed stronger aversion to the high-variance option because the cumulative probability of failing to reach the distant reward increased exponentially with depth. Moreover, this pattern held both for infinite rollouts and realistic, finite rollouts. With finite rollouts, the transition between preferences was simply more gradual, as choices depended on stochastic over- or under-sampling of the rewarding path (Figure 4B). These results confirm that uncertainty about reaching future states naturally produces temporal discounting.

We then tested if the same mechanism could explain preference for immediate gratification despite future costs, which reflects real-life dilemmas like procrastination or addiction. We formalized this with an action that yields an immediate reward of $+10$ with a larger punishment of -15 in the future. Here, uncertainty ϵ represented the agent's perceived probability of escaping from the inevitable future punishment at each step. The results showed a mirror image of our

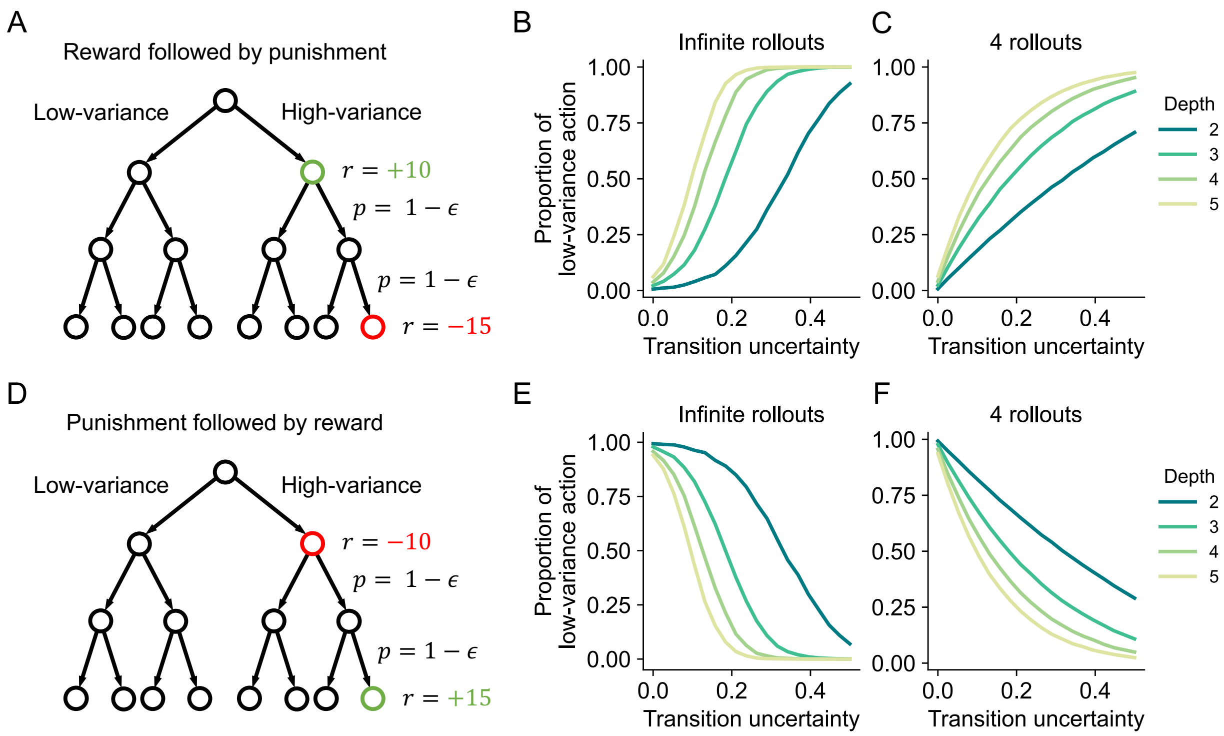

Figure 5: Temporal discounting can arise from rational rollouts under uncertainty. (A) Illustration of an environment with an early punishment for one action followed by a larger reward later in the tree. Transition uncertainty corresponds to the parameter ϵ that characterizes the probability of failing to transition towards the terminal leaf node with a large reward. (B-C) Increasing uncertainty favors the low-variance action, despite the high-variance action being optimal. (D) Illustration of an environment with an early reward for one action followed by a larger punishment later in the tree. Transition uncertainty corresponds to the parameter ϵ that characterizes the probability of escaping from the path towards the large punishment. (E-F) Increasing uncertainty favors the high-variance action, despite the low-variance action being optimal. Shaded regions show standard errors across participants.

previous findings: when uncertainty was high, the agent optimistically gambled on escaping the future punishment and chose the risky, high-variance action. As uncertainty decreased, the agent increasingly recognized the inevitability of the punishment and opted for the safer, low-variance alternative (Figure 5E-F). Taken together, these symmetric results show that apparent myopic choices can emerge directly from a rational evaluation of outcomes under uncertainty, without requiring any *a priori* devaluation of the future.

These findings demonstrate that a rollout-based agent can exhibit behaviors that resemble temporal discounting (Frederick et al., 2002; Mazur, 2013; Laibson, 1997; Green and Myerson, 2004). In fact, if the payout at all other leaf nodes is zero, our rollout model becomes mathematically equivalent to exponential discounting. However, the conceptual foundations are distinct. Classic discounting models assume that future rewards are intrinsically less valuable or that there is a probability of the entire task terminating (Frederick et al., 2002; Samuelson, 1937; Ainslie, 1975) . In contrast, our model values all rewards equally and assumes a fixed task horizon. Apparent discounting is not a built-in bias, but rather an emergent consequence of rational evaluation: as the agent plans further into the future, the uncertainty about which states it will reach increases, which effectively and rationally devalues distant outcomes. While our framework provides a rational basis for such discounting, it may not fully explain more complex phenomena like "pruning," a powerful heuristic where humans avoid branches with large punishments to a degree that requires a second, outcome-specific discount factor (Huys et al., 2012). We speculate that such effects could potentially be captured by extending our model to include mechanisms like increasing uncertainty with planning depth or by adopting more adaptive rollout strategies, such as Monte Carlo Tree Search.

2.5 Rollouts implement an evidence accumulation process

To connect our framework to established theories, we investigated how our rollout-based model relates to evidence accumulation models. Such models posit that decisions in bandit tasks emerge from accumulating noisy samples until

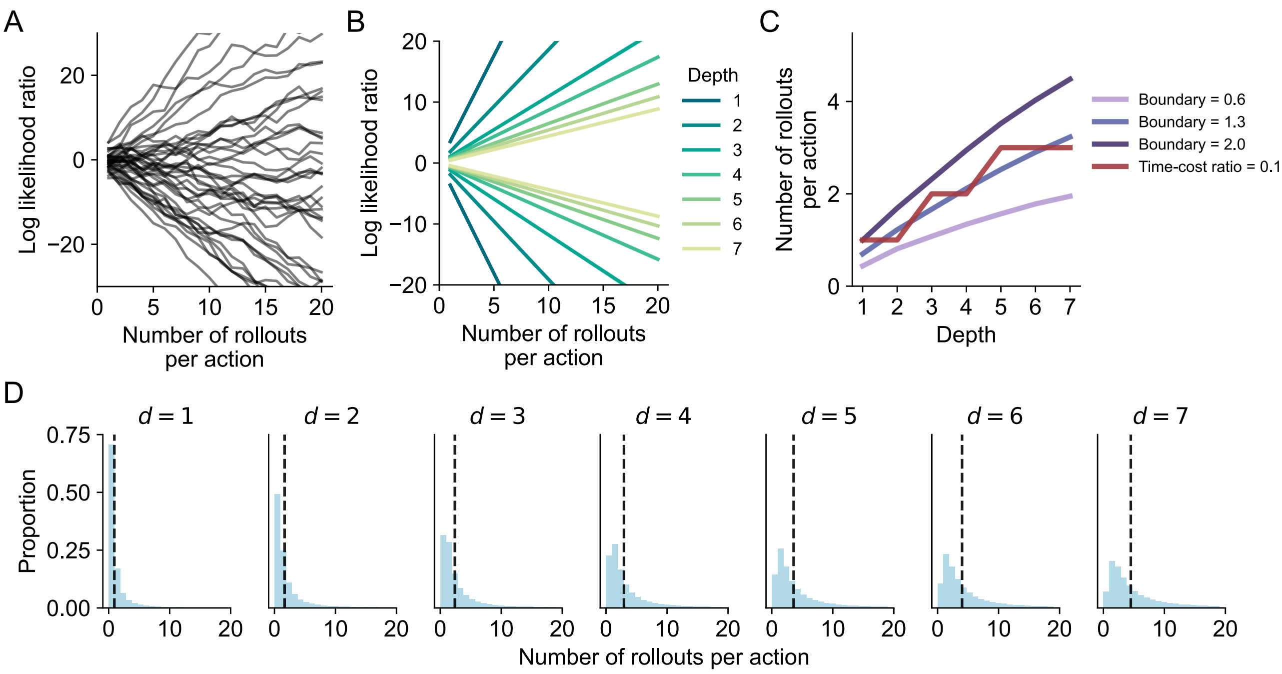

Figure 6: Planning with rollouts resembles evidence accumulation. (A) Evidence accumulation dynamics across rollouts in individual trials, illustrated with a tree depth of $d = 4$. (B) Evidence accumulation dynamics averaged across trials for different tree depths. Curves with positive and negative slopes correspond to the accumulation process under the two hypotheses, respectively. (C) Number of rollouts per action as a function of tree depth. Blue curves correspond to the predictions of the constant boundary model. The red curve corresponds to the predictions of the model that maximizes reward fraction per time under a time-cost ratio of 0.05. (D) The distribution of Shaded regions show standard errors.

reaching a decision boundary (Bakkour et al., 2018; Gold and Shadlen, 2007; Ratcliff and McKoon, 2008; Zylberberg et al., 2024). Recent work suggests that sequential decision-making employs similar accumulation processes (Solway and Botvinick, 2012). Here, our goal is to test whether our rollout model produces the characteristic dynamics of evidence accumulation and whether it makes testable predictions about how decision complexity affects reaction times.

To map rollouts onto evidence accumulation, we followed previous work (Vul et al., 2014) and framed the choice in a binary tree ($b = 2$) as a hypothesis testing problem: deciding whether action 1 or action 2 is superior. At each step, the agent samples one rollout for each action, computes the difference in their empirical cumulative rewards (Δg), and updates a log-likelihood ratio ($\mathcal{L}$) representing the accumulated evidence for one hypothesis over the other:

$$\mathcal{L} = \log \frac{p(\Delta g \mid \text{action 1 better})}{p(\Delta g \mid \text{action 2 better})}. \tag{3}$$

The agent sums these ratios across rollouts and selects the better-supported action.

Our simulations revealed that this process closely mirrors the dynamics of classic evidence accumulation models, with the agent integrating noisy evidence over successive rollouts toward a decision (Figure 6A). Critically, we found that the speed of this accumulation process depended on the structure of the decision tree. As tree depth increased, the average accumulation speed became slower (Figure 6B), requiring more rollouts to reach a decision boundary (Figure 6C). This was reflected in the distribution of rollout counts, which shifted higher and became broader for deeper trees (Figure 6D). These findings lead to a clear and testable prediction: humans should take longer to make decisions in problems that require deeper planning.

To understand why deeper trees slow down evidence accumulation, we analyzed how nodes at different depths contribute to the evidence. We found that early nodes, particularly the depth-1 nodes unique to each action, provide a consistent, deterministic "signal" to the cumulative reward. In contrast, deeper nodes, which are numerous and have a diminishing influence on the total reward, primarily add "noise". Their rewards, initialized symmetrically around zero, tend to cancel out in expectation but increase the overall variance of the cumulative reward. As a result, increasing tree depth leaves the signal (mean reward difference) largely unchanged while systematically increasing the noise (variance), thus lowering the signal-to-noise ratio (Figure S3). This mechanism closely resembles a drift-diffusion model, where early nodes supply the drift and deeper nodes supply the noise. This insight also bridges mechanistic and normative accounts of decision-making. We found that the number of rollouts predicted by a model with a fixed evidence boundary closely aligned with the optimal number of rollouts predicted by our rollout-based model, which maximizes reward per time without an explicit threshold (Figure 6C). This suggests that the amount of evidence humans gather is not arbitrary but is finely tuned to be efficient.

3 Discussion

In this work, we present a theoretical framework showing that effective decision-making in multi-step environments can be achieved with surprisingly few mental simulations, or rollouts. Building on insights from simpler choice paradigms (Vul et al., 2014), we demonstrate that deciding after minimal sampling is not a cognitive limitation but the optimal strategy under resource constraints. Our central contribution is to show that this efficiency arises from the correlated reward structure of naturalistic environments, which allows information from a single simulation to generalize across many related future paths. Together, these findings offer a normative foundation for the remarkable efficiency of human planning.

One of our key insights is that the efficiency of mental simulation is critically dependent on the structure of the environment. In the correlated environments we model, a single forward simulation informs not only the path it traverses but also other paths that share intermediate nodes. As a consequence, while the number of possible futures grows exponentially with planning depth, the optimal number of simulations needed for a good decision grows only linearly (Figure S2). An agent can thus exploit these statistical dependencies to make good decisions without exhaustively simulating every possibility. In contrast, this efficiency vanishes in uncorrelated environments where each path is independent, leading to a rapidly exploding demand for computational resources. This highlights reward correlation as a fundamental principle enabling sample-efficient planning.

Beyond explaining why few samples are sufficient, our framework offers a rational account for other well-known features of human planning, including apparent myopia and the dynamics of decision time. Traditional models explain temporal discounting by assuming that future outcomes are intrinsically devalued *a priori*. We show instead that myopic choices — such as avoiding paths with early punishments — can arise naturally from a rational evaluation under uncertainty, an effect that is amplified by limited sampling. Furthermore, our model connects directly to evidence accumulation frameworks, predicting that deeper decision trees should lead to slower decisions. We show this is because early nodes in a tree provide a stable "signal," while deeper nodes add "noise," reducing the signal-to-noise ratio and slowing accumulation. Impressively, the optimal number of rollouts predicted by our efficiency-based model closely matches the number predicted by a constant-confidence evidence accumulation model, suggesting a deep connection between resource rationality and cognitive heuristics.

For the sake of tractability, our model makes several simplifying assumptions, each of which presents a clear direction for future research. First, we assumed constant transition distributions beyond the root node, simplifying away the full sequential decision problem where agents optimize policies throughout the tree. While this captures many real scenarios — like investing where future market fluctuations remain beyond our control — it differs from settings where the policy can be updated at every node. Relaxing this assumption would likely increase the optimal rollout count, as deep rollouts would update policies at intermediate nodes, potentially invalidating earlier simulations. Second, we assumed simple, random rollouts as the mechanism for mental simulation. While rollouts are a core component of planning (Jensen et al., 2023; Olieslagers and Ma, 2025), humans likely employ more structured and adaptive search algorithms (Callaway et al., 2022; Huys et al., 2012). Examining the efficiency of these sophisticated strategies, such as Monte Carlo Tree Search or algorithms that incorporate depth-dependent costs, is a promising avenue for extending this work and accounting for a still broader range of human planning behaviors. Third, our experiments used explicit visual trees with random rewards, which may not fully capture mental representations of naturalistic problems with predictable reward patterns. Despite these limitations, our framework's core insight—that correlation enables efficient generalization—should extend to these richer settings.

In summary, we provide a computational framework for understanding how effective decisions in multi-step environments can be achieved with few rollouts. Our work demonstrates that the efficiency of human planning can be understood as a rational adaptation to the statistical structure of the world. This perspective opens up new possibilities for unifying computationally intensive search algorithms from machine learning with the elegant, resource-rational heuristics of human cognition.

4 Methods

4.1 Model

4.1.1 A decision tree environment

We model the environment as a decision tree with a branching factor b and a depth of d. Each node s in the tree is associated with a stochastic reward r_s, drawn from a Gaussian distribution:

$$r_s \sim \mathcal{N}(\mu_s, \sigma_r^2), \tag{4}$$

where μ_s is the true underlying expected cumulative reward at node s and is not directly observable. We assume that μ_s is itself independently initialized from a zero-mean Gaussian distribution:

$$\mu_s \sim \mathcal{N}(0, \sigma_\mu^2). \tag{5}$$

This two-level generative process reflects both uncertainty in individual observations (via σ_r^2) and stochasticity in the underlying reward landscape (via σ_μ^2).

Let $\mathcal{P}(s_i^{\text{leaf}})$ be the set of nodes along the unique path from the root node to a leaf node s_i^{leaf} (hereafter denoted as s_i for brevity). The empirical cumulative reward g_i for reaching the leaf node s_i is given by the sum of the rewards encountered along this path:

$$g_i = \sum_{s \in \mathcal{P}(s_i)} r_s. \tag{6}$$

Since the reward at different nodes are independent Gaussian variables, the empirical cumulative reward g_i therefore also follows a Gaussian distribution:

$$g_i \sim \mathcal{N}(m_i, \sigma_g^2), \tag{7}$$

where the mean and variance are given by:

$$m_i = \sum_{s \in \mathcal{P}(s_i)} \mu_s, \tag{8}$$

$$\sigma_g^2 = d\sigma_r^2. \tag{9}$$

In this formulation, the expected cumulative reward m_i is a sum of the unobserved local means, and the uncertainty (variance) σ_g grows linearly with tree depth due to the additive nature of independent uncertainty.

4.1.2 Reward correlation structure

A key feature of the decision tree environment is its reward correlation between different paths. Specifically, leaf nodes that are closer to each other in the tree share more early common nodes, and their cumulative rewards tend to be more correlated. This arises naturally from the tree structure: the more two paths overlap, the more similar their reward histories become.

To formalize this intuition, we examine the covariance of the expected cumulative rewards between different paths. Since the expected cumulative reward of a leaf node s_i (denoted as m_i) is a sum over independent Gaussian variables, we have:

$$\mathbb{E}[m_i] = 0, \tag{10}$$

$$\mathrm{Var}[m_i] = d\sigma_\mu^2. \tag{11}$$

Now, consider two leaf nodes s_i and s_j. Let s_{ij}^{MRCA} denote their most recent common ancestor (MRCA), which is the deepest node that lies on both paths from the root to s_i and s_j. The depth of this MRCA is d_{ij}^{MRCA}. Specially, when $s_i = s_j$, we have $s_{ij}^{\text{MRCA}} = s_i = s_j$ and $d_{ij}^{\text{MRCA}} = d$. Because the two paths $\mathcal{P}(s_i)$ and $\mathcal{P}(s_j)$ share exactly the first d_{ij}^{MRCA} nodes, the covariance between their expected cumulative rewards arises only from these shared components:

$$\mathrm{Cov}(m_i, m_j) = d_{ij}^{\text{MRCA}} \sigma_\mu^2. \tag{12}$$

We can collect the expected cumulative rewards of all $L = b^d$ leaf-node paths into a vector $\mathbf{m}$ and describe its distribution as a multivariate Gaussian:

$$\mathbf{m} \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}), \tag{13}$$

where the covariance matrix $\mathbf{\Sigma} \in \mathbb{R}^{L \times L}$ is defined elementwise as:

$$\Sigma_{ij} = d_{ij}^{\mathrm{MRCA}} \sigma_\mu^2. \tag{14}$$

This covariance structure encodes a hierarchical reward similarity among all leaf-node paths in the decision tree. Specifically, the covariance of the expected cumulative rewards between two paths depends on the depth of their MRCA. The more two paths overlap, the larger d_{ij}^{MRCA} is, and the more similar their expected cumulative rewards become. This property mirrors many real-world scenarios, where identical early decisions (e.g., strategic choices or initial conditions) often lead to similar future outcomes.

4.1.3 The informativeness of a single rollout

To formally quantify the informativeness of a single rollout, we consider how a single observation updates the agent's beliefs about the expected cumulative rewards of all paths. Let

$$\mathbf{m}^{(t)} \sim \mathcal{N}(\boldsymbol{\nu}^{(t)}, \mathbf{\Sigma}^{(t)}), \tag{15}$$

denote the latent (unobserved) vector of expected cumulative rewards of all $L = b^d$ leaf-node paths at time step t. Here, $\boldsymbol{\nu}^{(t)}$ and $\mathbf{\Sigma}^{(t)}$ are the agent's current belief mean and covariance. Specifically, we assume that the belief at the initial time step is given by the prior defined in Equation (13), such that $\boldsymbol{\nu}^{(0)} = \mathbf{0}$ and $\mathbf{\Sigma}^{(0)} = \mathbf{\Sigma}$.

The agent performs a rollout that terminates at leaf node s_i, yielding a noisy observation of the cumulative reward g_i along that path. Conditional on a particular realization of the latent state $\mathbf{m}^{(t)}$, this observation is drawn from

$$g_i \mid \mathbf{m}^{(t)} \sim \mathcal{N}(m_i^{(t)},\, d\sigma_r^2), \tag{16}$$

where $m_i^{(t)}$ denotes the i-th component of the latent vector $\mathbf{m}^{(t)}$, i.e., the true (unobserved) expected cumulative reward for path i at time t.

Using standard results for conditioning a multivariate Gaussian, the posterior over the expected cumulative rewards after observing g_i is given by:

$$\mathbf{m}^{(t+1)} \mid g_i \sim \mathcal{N}(\boldsymbol{\nu}^{(t+1)},\, \mathbf{\Sigma}^{(t+1)}), \tag{17}$$

$$\boldsymbol{\nu}^{(t+1)} = \boldsymbol{\nu}^{(t)} + \frac{\mathbf{\Sigma}_{\cdot i}^{(t)}}{\Sigma_{ii}^{(t)} + d\sigma_r^2}(g_i - \nu_i^{(t)}), \tag{18}$$

$$\mathbf{\Sigma}^{(t+1)} = \mathbf{\Sigma}^{(t)} - \frac{\mathbf{\Sigma}_{\cdot i}^{(t)} \mathbf{\Sigma}_{\cdot i}^{(t)\top}}{\Sigma_{ii}^{(t)} + d\sigma_r^2}. \tag{19}$$

where $\mathbf{\Sigma}_{\cdot i}^{(t)}$ denotes the i-th column of the covariance matrix $\mathbf{\Sigma}^{(t)}$.

To gain insight into its implications, we consider the posterior distribution over the expected cumulative reward for another leaf node s_j after the first rollout, given the observation g_{s_i}. This is a one-dimensional marginal from the multivariate posterior:

$$m_j^{(1)} \mid g_i \sim \mathcal{N}(\nu_j^{(1)}, \Sigma_{ij}^{(1)}), \tag{20}$$

$$\nu_j^{(1)} = \frac{d_{ij}^{\mathrm{MRCA}} \sigma_\mu^2}{d(\sigma_\mu^2 + \sigma_r^2)} g_i \tag{21}$$

$$\Sigma_{jj}^{(1)} = \sigma_\mu^2 - \frac{(d_{ij}^{\mathrm{MRCA}} \sigma_\mu^2)^2}{d(\sigma_\mu^2 + \sigma_r^2)}. \tag{22}$$

This expression reveals several important intuitions:

- Posterior mean: The updated estimate of the expected cumulative reward for s_j is proportional to the observed value g_i, scaled by a factor that depends on the depth of the MRCA between nodes s_i and s_j. If s_i and s_j share more of their path (i.e., if d_{ij}^{MRCA} is large), then the information from the rollout to s_i is highly informative for estimating m_j. In contrast, if the paths diverge early (i.e., if d_{ij}^{MRCA} is small), the posterior mean for m_j remains close to zero, indicating little information transfer.
- Posterior variance: The uncertainty about m_j also depends on the depth of the MRCA. The more overlap the two paths are, the more the uncertainty about m_j is reduced. As $d_{ij}^{\text{MRCA}} \to d$, the posterior variance becomes closer to zero, meaning we nearly recover the true mean. Conversely, when $d_{ij}^{\text{MRCA}} \to 0$, the posterior variance stays close to the prior variance σ_μ^2, indicating that little has been learned.

After the first rollout, although the posterior mean and variance becomes more complex, the MRCA structure continues to shape the updates. Equation (19) shows that each new rollout updates beliefs about all paths in proportion to their current covariance with the sampled path. Since the initial covariance matrix is defined by MRCA depth and each posterior covariance is constructed via rank-one reductions of this matrix, the hierarchical correlation structure is preserved throughout. The effect of each rollout continues to propagate more strongly to paths that share earlier ancestors with the sampled one. Therefore, while only the first rollout yields a closed-form posterior in terms of MRCA depth, the recursive update guarantees that this same intuition holds across multiple rollouts.

In summary, the informativeness of a single rollout generalizes to other paths in proportion to the overlap. This hierarchical generalization property is a direct consequence of the tree's reward correlation structure and plays a central role in the efficiency of rollout-based decision-making.

4.1.4 Agent

We consider a decision-making scenario where an agent must select an action at the root node of a decision tree in order to maximize the expected cumulative reward. While the root-node policy is thus controlled by the agent, we assume that all subsequent transitions are governed by fixed, stochastic dynamics. The agent thus faces the challenge of evaluating the long-term consequences of its own actions. We assume the agent is equipped with a world model consisting of reward and transition probabilities. The agent can use this model to simulate possible outcomes by drawing rollouts—sampled trajectories from the root to a leaf node. Each rollout provides a noisy estimate of the expected cumulative reward for the corresponding root-node action. Formally, the agent chooses a root-node action a and performs a rollout. This rollout terminates at a leaf node s_i, producing an observed cumulative reward g sampled from the distribution $p(g \mid a)$, which is defined as:

$$p(g|a) = \sum_{s_i} p(g|s_i)p(s_i|a). \tag{23}$$

Here, $p(g|a)$ is the empirical cumulative reward distribution conditioned on choosing the root-node action a for a rollout, marginalized across all leaf nodes s_i that the rollout could visit.

Prior to selecting an action, the agent samples a finite number of rollouts from its world model. Each rollout corresponds to a root-node action followed by a stochastic trajectory to a leaf node and yields one cumulative reward sample. Let T denote the full set of sampled rollouts. We assume that the agent samples rollouts using a rollout policy that sequentially samples each action. This sequential rollout policy assumes no systematic selection of root-node actions and samples all actions as evenly as possible. In the simplest case, we further assume that the transition probabilities beyond the root node are uniform. This implies that for any given root-node action, all downstream leaf nodes are equally likely to be visited.

After sampling the rollouts, the agent takes the action with the highest average cumulative reward estimated from its rollouts:

$$a^* = \underset{a}{\operatorname{argmax}} \left[\frac{1}{|T_a|} \sum_{g \in T_a} g \right], \tag{24}$$

where $T_a \subseteq T$ is the subset of rollouts from the root-node action a, and g is the empirical cumulative reward of the sampled rollout.

4.2 Simulation

We simulated our agent with a series of parameter combinations. Across all simulations, we set the variance of the reward mean $\sigma_\mu^2 = 1$, and the variance of the observational noise $\sigma_r^2 = 0.3$. Here we detail the settings for each simulation specifically.

4.2.1 Optimal decisions based on few rollouts (Figure 2)

In the simulation, transition probabilities beyond the root node were uniform, and rollouts always terminated at leaf nodes. Each rollout had a cost of 0.1 (i.e., one action took as much time as 10 rollouts). We simulated the agent with tree depth $d \in \{1, 2, \cdots, 7\}$, branching factor $b = 2$, number of rollouts $N^{\text{rollout}} \in \{0, 1, 2, \cdots, 20\}$. For each parameter combination, we averaged the choices over 1000 environments with 1000 trials per environment.

4.2.2 Optimal simulation depth (Figure 3)

In the simulation, transition probabilities beyond the root node were uniform. Rollouts could end at intermediate nodes, limiting the agent's access to information to a specified depth, while performance was still evaluated with respect to the rewards over the entire tree. We simulated the agent with tree depth $d = 5$, branching factor $b \in \{2, 3, 4\}$, and computational budget (number of node expansions) $N^{\text{budget}} \in \{0, 1, 2, \cdots, 120\}$. For each parameter combination, we averaged the choices over 1000 environments with 1000 trials per environment.

4.2.3 Temporal discounting (Figure 4)

In the simulation, environment transitions were deterministic and the agent had full control over its policy at every node. One action was configured to yield a punishment of -10 (or a reward of $+10$) at depth 1, followed by a reward of $+15$ (or a punishment of -15) at a downstream leaf node, while all other nodes contained rewards drawn from the standard distribution. To model uncertainty in the world model, we introduced a perceived failure probability ϵ for each intended transition toward the large reward or punishment. Rollouts always terminated at leaf nodes. We simulated the agent with tree depth $d \in \{2, 3, 4, 5\}$, branching factor $b = 2$, number of rollouts $N^{\text{rollout}} \in \{4, 20, \infty\}$, and transition uncertainty $\epsilon \in [0, 0.5]$. The ground-truth action values were used when $N^{\text{rollout}} = \infty$. We increased the number of sampled environments and testing trials, since under uneven transition probabilities, the resulting value estimations were more noisy. Specifically, for each parameter combination, we sampled 2000 environments with 5000 trials per environment.

4.2.4 Relationship with evidence accumulation (Figure 5)

In the simulation, transition probabilities beyond the root node were uniform, and rollouts always terminated at leaf nodes. At each time step, the agent sampled one rollout per action, computed the difference in their empirical cumulative rewards, updated a log-likelihood ratio representing the accumulated evidence, and summed these ratios across rollouts before selecting the better-supported action. We simulated the agents with tree depth $d \in \{1, 2, \cdots, 7\}$, branching factor $b = 2$, number of rollouts $N^{\text{rollout}} \in \{0, 1, 2, \cdots, 20\}$. We compared two agent types: one using a fixed decision boundary and one maximizing reward per time without an explicit threshold. For the agent with a fixed decision boundary, we simulated the model with decision boundaries including 0.6, 1.3, and 2, while for the agent maximizing reward per time, we simulated the model with a cost per rollout of 0.1. To estimate log-likelihoods, we first sampled 1000 environments, each with 1000 testing trials. We aggregated these samples to obtain the empirical distribution of cumulative reward differences between the two root-node actions. We then approximated this distribution with a Gaussian distribution by using the empirical mean and standard deviation as the parameters of the Gaussian distribution. The estimated Gaussian distribution allowed us to compute the log-likelihood of individual rollouts. We repeated this procedure across tree depths. To generate Figure 5 and Figure 5B, we averaged the log-likelihood ratios over 2000 environments and 2000 testing trials per environment.

4.3 Experiment

4.3.1 Participants

We recruited participants on Prolific and excluded participants whose accuracy was below chance level ($1/b$). For Experiment 1, 50 participants were recruited and 3 were excluded, resulting in a final sample of 47 participants. For Experiment 2, 60 participants were recruited and were randomly assigned to one of the three conditions ($[b = 2, d = 3]$, $[b = 3, d = 3]$, and $[b = 2, d = 4]$) such that each condition had 20 participants. 5 participants were excluded, leading to a final sample of 55 participants (20 in $[b = 2, d = 3]$, 17 in $[b = 3, d = 3]$, and 18 in $[b = 2, d = 4]$).

4.3.2 Task

In the experiments, participants interacted with a visually displayed decision tree. The tree consisted of 31 nodes (or 40 nodes in condition $b = 3, d = 3$ in Experiment 2). Nodes were arranged evenly on a circle. Each node was associated with a reward sampled from the set of integers from -8 to 8, with probabilities proportional to a Gaussian distribution with a mean 0 and a standard deviation 10. Rewards were indicated by color, with yellow representing high rewards and blue representing low rewards, and the transitions were shown with arrows. The transition and reward structure was sampled independently on every trial to ensure that participants could memorize the tree structure but only used decision-time planning to solve the task. In Experiment 1, the tree had a branching factor of $b = 2$, and the tree depth was uniformly sampled from 1 to 4. In Experiment 2, participants were assigned to one of the three conditions: $[b = 2, d = 3]$, $[b = 3, d = 3]$, or $[b = 2, d = 4]$. Before beginning the main experiment, participants went through an interactive instruction phase that explained rules of the task, followed by 3 practice trials.

Before selecting an action, participants could sample rollouts to collect information. At the start of each trial, only the root node (marked in red) and its two outgoing arrows were visible. Participants could hover their mouse over one of the children (nodes pointed to by the arrows) to reveal its reward and the transitions to its children. Once they hover over a node, the information revealed by the previous hover was again hidden. Participants could also hover over the root at any time, which reset the revealed information to the starting state. This design allowed participants to sample rollouts, i.e., sequences of node reveals starting from the root and terminating at an arbitrary node.

In Experiment 1, we imposed a time limit for the rollout phrase. Time limits were sampled from a log-uniform distribution between 1 and 21 seconds, then shifted by -1 to produce possible time limits between 0 and 20 seconds. Participants were not informed of the time limit of the trial so that they could not adjust their rollout strategies based on the time left. In Experiment 2, we instead imposed budget on the number of node expansions. The budgets were sampled from a log-uniform distribution over integers between 4 and 23, then shifted by -3 to yield budgets between 1 and 20 node expansions. Hovering over the root did not consume the budget, allowing participants to reset for free. Participants were informed informed of the remaining budget of the trial, so they could adapt their rollout depth accordingly. Importantly, in both experiments, participants were not allowed to end the rollout phase early or to continue once the time or budget limit was reached. This ensures control over the number of simulations performed by participants.

After the rollout phase, participants were required to select a root-node action by clicking on one of the depth-1 nodes. The remainder of the trajectory was then sampled randomly. Nodes along the trajectory lit up sequentially, and their associated reward values were revealed along with their integer labels to provide feedback. Participants completed 100 trials in total for one experiment.

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5 Supplementary Information

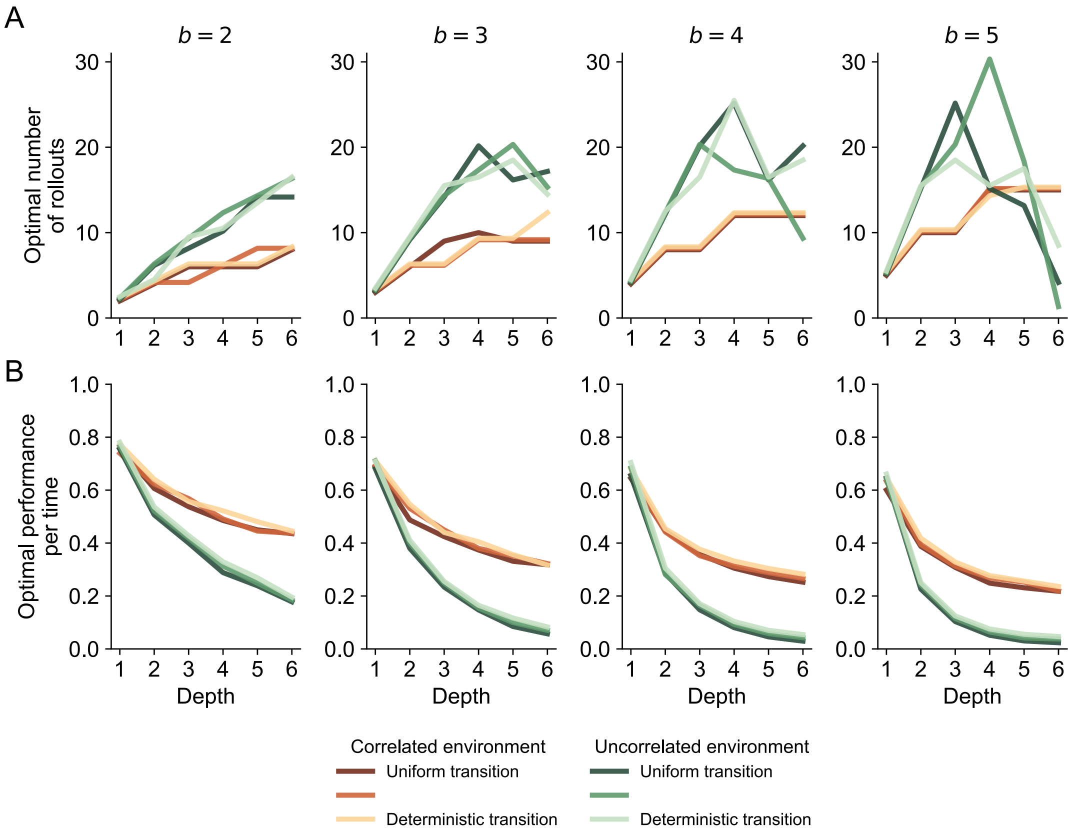

Figure S1: Model performance under various conditions. (A) Optimal number of rollouts as a function of tree depth under different branching factors, transition probabilities, and correlation structures. (B) Performance per time achieved by the agent under the optimal number of rollouts. In uncorrelated environments, performance per time drops quickly with increasing tree depth or branching factor, resulting in a noisy and unstable estimate of the optimal number of rollouts. For each parameter combination, choices are averaged over 1000 environments, each with 1000 trials.

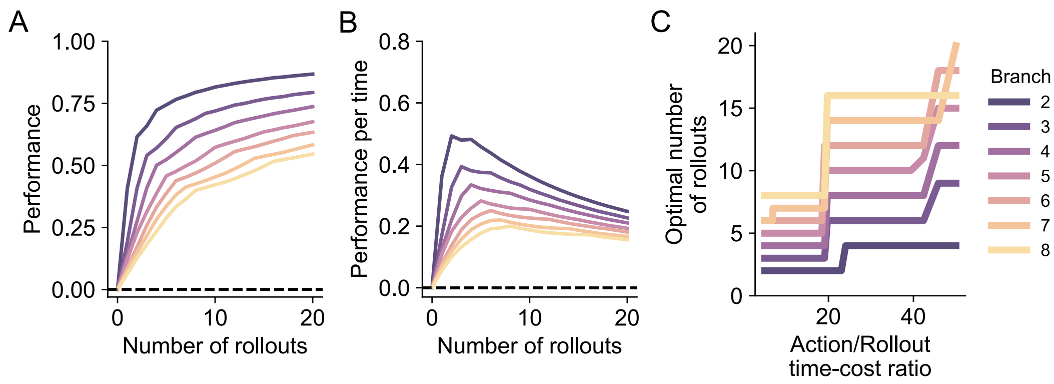

Figure S2: Optimal decisions based on few rollouts under different branching factors. (A) Performance achieved as a function of the number of rollouts performed before taking an action. (B) Performance achieved divided by the time needed to execute all the rollouts and one action, as a function of the number of rollouts performed before taking the action. (C) Optimal number of rollouts that yield the highest performance per time as a function of action/rollout time-cost ratio. Here we illustrate with a tree depth $d = 2$. For each parameter combination, choices are averaged over 1000 environments, each with 1000 trials.

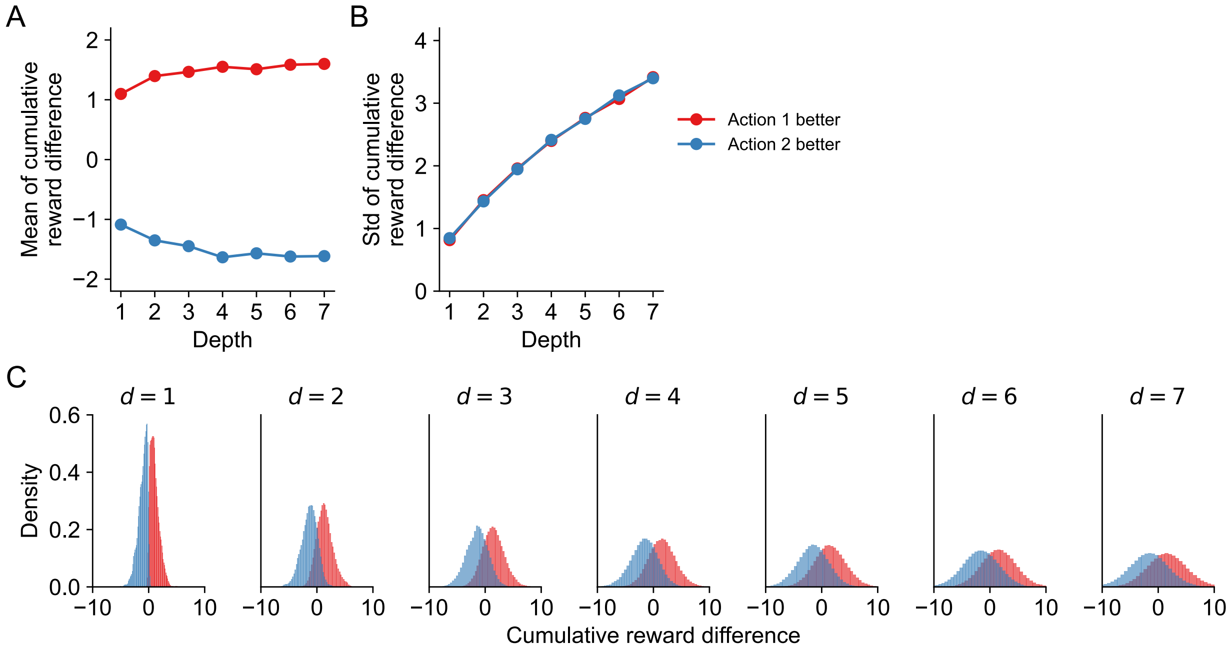

Figure S3: Cumulative reward difference. Mean (A), standard deviation (B) and the raw distribution (C) of cumulative reward difference for the two hypotheses (action 1 better and action 2 better) under different tree depths. For each depth, results are aggregated over 2000 environments, with 500 sampled rollouts per action in each environment.